THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — MATHS TWO-CHAPTER TEST SET C

Chapters 9 & 10: **The Baudhayana–Pythagoras Theorem, Proportional Reasoning-2** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat figures wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The legs of a right triangle are 7 cm and 24 cm. Its hypotenuse is: 1

(a) 31 cm **(b)** 26 cm **(c)** 24 cm **(d)** 25 cm

2. The legs of a right triangle are 12 cm and 16 cm. Its hypotenuse is: 1

(a) 19 cm **(b)** 20 cm **(c)** 28 cm **(d)** 21 cm

3. The hypotenuse of a right triangle is 25 cm and one leg is 7 cm. The other leg is: 1

(a) 23 cm **(b)** 25 cm **(c)** 18 cm **(d)** 24 cm

4. Is a triangle with sides 9, 12 and 16 a right triangle? 1

(a) Cannot say **(b)** Yes

(c) Only if it is isosceles **(d)** No

5. Is a triangle with sides 10, 11 and 15 a right triangle? 1

(a) Cannot say **(b)** Only if it is isosceles

(c) No **(d)** Yes

6. On a map, 1 cm represents 12.5 km. A distance of 8 cm on the map represents: 1

(a) 50 km **(b)** 100 km **(c)** 112.5 km **(d)** 200 km

7. ₹ 1100 is shared among A, B and C in the ratio 2:3:6. B gets: 1

(a) ₹ 200 **(b)** ₹ 600 **(c)** ₹ 366 **(d)** ₹ 300

8. ₹ 720 is shared among A, B and C in the ratio 1:3:5. B gets: 1

(a) ₹ 240 **(b)** ₹ 80 **(c)** ₹ 320 **(d)** ₹ 400

9. 15 workers finish a job in 14 days. How many days will 10 workers take? 1

(a) 20 **(b)** 9.33333 **(c)** 21 **(d)** 23

10. In a pie chart, a sector representing 10% of the data has a central angle of: 1

(a) 10° **(b)** 36° **(c)** 20° **(d)** 46°

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 10–15 words each.

11. Is the triangle with sides 5 cm, 6 cm and 8 cm right-angled? 2

12. The sides of a rectangle are 20 m and 21 m. Find its diagonal. 2

13. 20 students eat in 12 days with a stock of food. How long does it last for 30 students? 2

14. State the difference between direct and inverse proportion. 2

SECTION — C (Short Answer Type) 4 × 3 = 12

Answer in about 30–40 words each.

15. A flagpole is supported by a 15 m wire fixed 9 m from its foot. Find the height at which the wire is attached. 3

16. A map scale shows 5 cm = 20 km. What is 1 cm in km, and what is the representative ratio? 3

17. Find the perimeter of a rhombus whose diagonals are 16 cm and 30 cm. 3

OR

The sides of a triangle are 15 cm, 20 cm and 25 cm. Show it is right-angled and find its area. 3

18. A farmer has fodder for 30 cows for 24 days. How long for 36 cows? 3

OR

A pie chart of 360 students shows: Maths 25%, Science 35%, Arts 40%. Find the angles and the number of students for each. 3

SECTION — D (Long Answer Type) 2 × 5 = 10

Answer in about 80–100 words each.

19. (i) Show that the sum of the squares of the diagonals of a rhombus equals four times the square of its side.
(ii) Verify for 6 and 8. 5

OR

Case study: A rectangular field is 48 m long and 14 m wide. 5
(i) Find the diagonal path.
(ii) Cost of laying a path along it at ₹ 40/m.
(iii) Compare with walking along the two sides.

20. (i) A cyclist covers 36 km in 2 hours. How long for 90 km?
(ii) What if the speed becomes 12 km/h? 5

OR

(i) If 15 men take 6 days, find the work in man-days. 5
(ii) How many men finish half the job in 3 days?

END OF QUESTION PAPER Total: 40 Marks

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